Aerospace Manufacturing Core
Study Guide

"AMCSG-301"

Blueprints & Documents

Introduction to Blueprints

Introduction to Blueprints

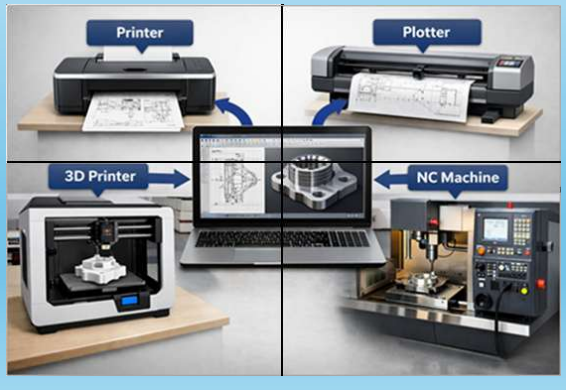
Text from Google AI

Definition of a Blueprint

- Blueprint = Engineering Drawings = Technical Drawings = Plans (= Sketch)
- Detailed drawings used all stages: design > manufacture > maintain aircraft
- Modern drawings = CAD. Blueprint comes from old chemical process for copies

(PD) U.S. Gov – NAVEDTRA14040A, 2015

- CAD drawings can then feed:
  - Printers
  - Plotters
  - 3D Printers
  - NC Machines



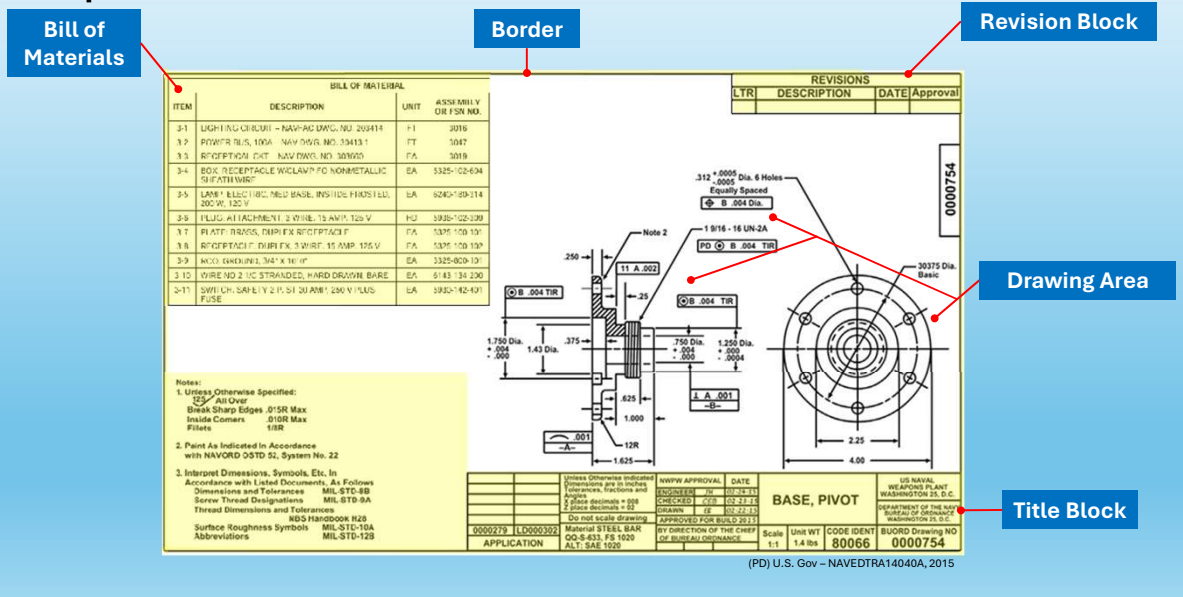
(PD) U.S. Gov – NAVEDTRA14040A, 2015

- Google AI: “show me a picture of blueprints in the aerospace industry. describe all parts of a blueprint.”

- US Govt Public Domain Doc Pictures: https://media.defense.gov/2017/Nov/03/2002658619/-1/-1/1/171103-N-ZY182-15347.pdf

- ChatGPT: make me a picture showing a laptop with CAD that feeds data to 1) printer, 2) plotter, 3) 3D printer to create part, 4) NC Machine / direct numerical control

Blueprint “Standard Elements” (1 of 2)



(PD) U.S. Gov – NAVEDTRA14040A, 2015

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Blueprint "Standard Elements" (2 of 2)

- **Blueprint Lines**
  - Shape of objects from top, front, and side views
  - Features hidden inside an object
  - Measurement and callout lines
- **Blueprint Symbols**
  - Hundreds of specific symbols
  - Representing Machining, Fasteners, Electrical, Hydraulic, Welding, etc.

- ChatGPT query identified USGovernment public domain document: https://media.defense.gov/2017/Nov/03/2002658619/-1/-1/1/171103-N-ZY182-15347.pdf

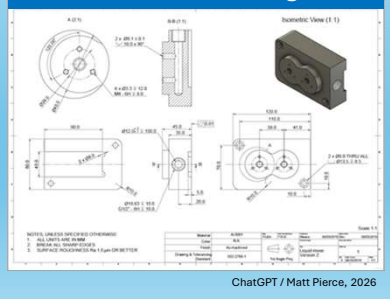
Text from ChatGPT

Drawing Types

- **Production Drawings** - Engineers' Instructions how to build
  1. **Detail Drawings** - How to fabricate a specific **part**
  2. **Assembly Drawings** - How an **assembly** is put together
  3. **Installation Drawings** - Where parts & assemblies located in **final product**

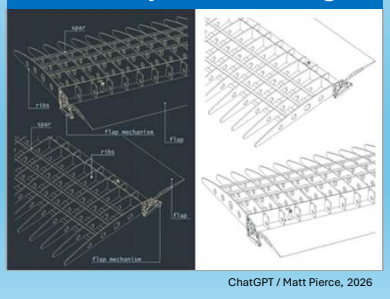
Part → Assembly → Final Product

Part **Detail Drawing**



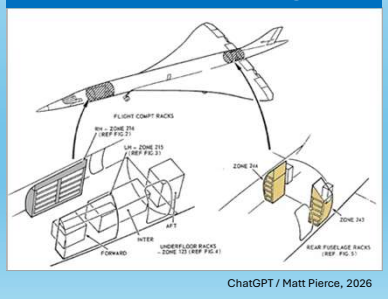
ChatGPT / Matt Pierce, 2026

Assembly of Parts **Drawing**



ChatGPT / Matt Pierce, 2026

Installation Drawing



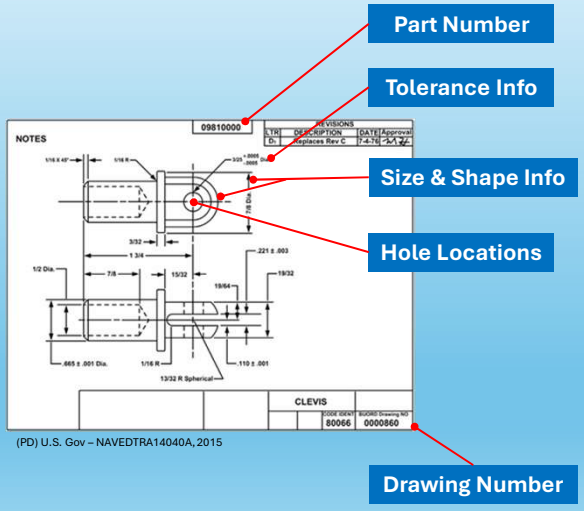
ChatGPT / Matt Pierce, 2026

- ChatGPT for all three images: show an example each of "Detail Drawings", "Assembly Drawings", and "Installation Drawings". Focus on commercial aircraft as the context.

Text from ChatGPT

“Detail” Drawings

- **“Part Details”** Drawing
- Material Size Requirements
- Heat-Treating Information
- Machine Finish Requirements
- Protective Finish Requirements
- Process Specs & Standards



(PD) U.S. Gov – NAVEDTRA14040A, 2015

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Text from ChatGPT

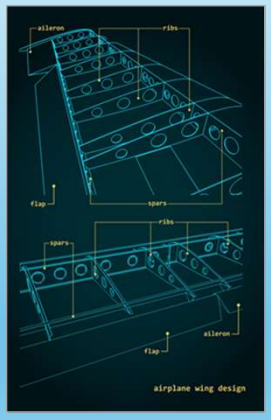
“Assembly” Drawings

- **“Assembly of Parts”** Drawing
- Part List
- Information How Parts Fit Together
- Part Number and Instructions
- Location dimensions
- Fastening Methods

BILL OF MATERIAL

ITEM	DESCRIPTION	UNIT	ASSEMBLY OR FSN NO.	QUANTITIES	
				TROP	NORTH
3-1	LIGHTING CIRCUIT – NAVFAC DWG. NO. 203414	FT	3076	3	16
3-2	POWER BUS, 100A – NAV DWG. NO. 30413 1	FT	3047	1	20
3-3	RECEPTICAL CKT – NAV DWG. NO. 303660	EA	3019	2	1
3-4	BOX, RECEPTACLE W/CLAMP F/O NONMETALLIC SHEATH WIRE	EA	5325-102-604	3	1
3-5	LAMP, ELECTRIC, MED BASE, INSTIDE FROSTED, 200 W, 120 V	EA	6240-180-314	60	6
3-6	PLUG, ATTACHMENT, 3 WIRE, 15 AMP, 125 V	HD	5936-102-309	10	1
3-7	PLATE: BRASS, DUPLEX RECEPTACLE	EA	5325-100-101	5	6
3-8	RECEPTACLE, DUPLEX, 3 WIRE, 15 AMP, 125 V	EA	5325-100-102	5	1
3-9	RCO, GROUND, 3/4" X 10' 0"	EA	3325-800-101	12	1
3-10	WIRE NO 2 1/C STRANDED, HARD DRAWN, BARE	EA	6143-134-200	52	1
3-11	SWITCH, SAFETY 2 P, ST 30 AMP, 250 V PLUS FUSE	EA	5930-142-401	2	1
3-12	CLAMP, GROUND ROD	EA	5008-100-101	13	13
3-13	SWITCH, SAFETY, 200 AMP, 250 V, 3 P	EA	6930-201-903	1	1
3-14	FUSE, RENEWABLE 200 AMP, 250 V	EA	6920-100-000	6	6
3-15	LINK FUSE, 200 AMP, 250 V	EA	6920-100-001	6	6
3-16	FUSE PLUG, 30 AMP, 125 V	EA	6920-100-102	12	12

(PD) U.S. Gov – NAVEDTRA14040A, 2015



ChatGPT / Matt Pierce, 2026

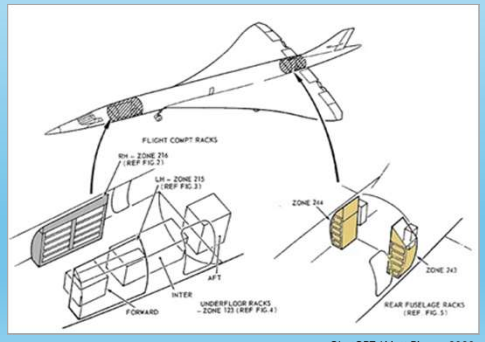
- ChatGPT: show examples of assembly drawings for aircraft manufacturing. include a part list, diagram how parts fit together, dimension lines, etc.

- ChatGPT query identified USGovernment public domain document: https://media.defense.gov/2017/Nov/03/2002658619/-1/-1/1/171103-N-ZY182-15347.pdf

Text from Google AI

“Installation” Drawings

- **“Final Installation of Assemblies and Parts”** Drawing
- List of Required Component Parts



ChatGPT / Matt Pierce, 2026

Q&A

Practice Q&A's (Q1-Q8)

Q1: General name covering the engineering drawings, technical drawing, or plans?

A: Blueprints

Q2: Modern blueprints are no longer hand drawn…what is the name of the software used?

A: CAD: Computer Aided Drafting

Q3: Both CAD and blueprints enabled duplicating drawings. But what more can CAD do?

A1: Send to 3D printer to create a part.

A2: Send to NC machine to mill a part.

Q4: Name of blueprint section listing materials and parts used on the drawing?

A: Bill of Materials

Q5: Name of blueprint section listing changes made, by whom, etc.?

A: Revision Block

Q6: Name of blueprint section listing the drawing number, description, notes, scale, etc.?

A: Title Block

Q7: Name of blueprint section containing schematics, drawings, callouts, dimensions, etc.?

A: Drawing Area

Q8: Name encompassing the Details Drawings, and the Assembly Drawings, and the Installation Drawings?

A: Production Drawings

Practice Q&A's (Q9-Q8)

Q9: Type of drawing showing how to fabricate a specific part?

A: Detail Drawing

Q10: Type of drawing showing how to assemble the parts into a component?

A: Assembly Drawing

Q11: Type of drawing showing where parts and assemblies go in final product?

A: Installation Drawing

Q12: What type of drawing contains part list, part numbers, location dimensions, and fastening methods?

A: Assembly Drawings

Q13: When you put together Ikea furniture by following the instruction manual, what is it?

A: Installation Drawing because parts already created, and sub-assemblies like latches and hinges already built. You are just installing it all together.

Q14: What type of drawing contains material size requirements, heat-treating info, and machine finish requirements?

A: Detail Drawings

Thank You

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https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg